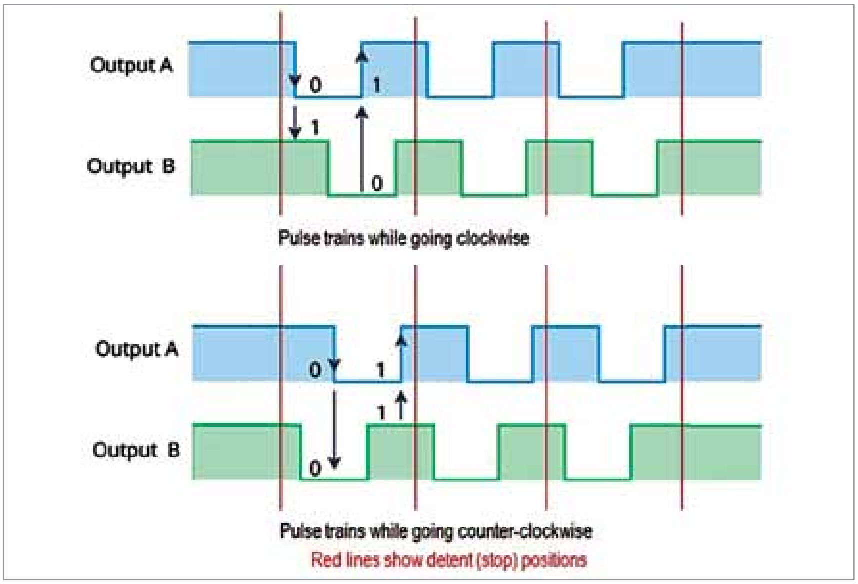

Fig. 6: Encoder switch output pulse trains

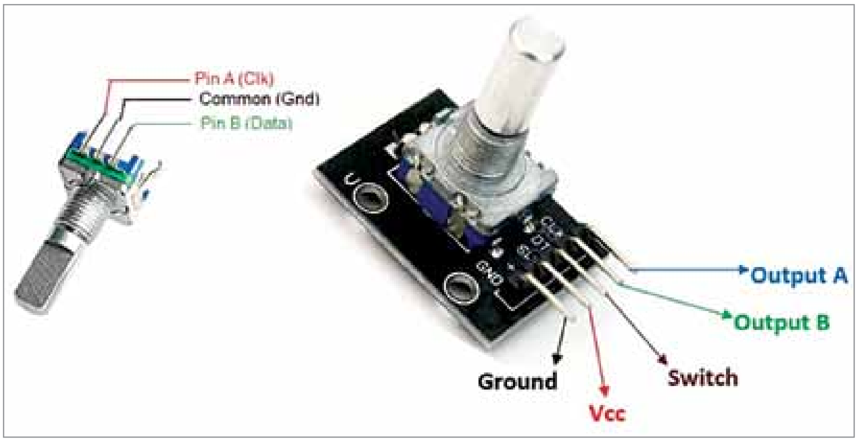

Fig. 7: Types of encoders

Fig. 8: Installation of the reed switches

Software

The software program *UVTimer.ino* can decode the direction of rotation of encoder just by checking the level of pin B when pin A changes from high to low. Pin B would be 1 if it goes clockwise and 0 if it goes counter-clockwise. A variable is saved, which keeps track of the number of pulses and reflects the new value after rotation. The initial value has to be given (0 in our case). Since the encoders are mechanical, there is a lot of bounce when position changes, requiring debouncing in software.

All the above tasks are handled by NewEncoder library. Pins A and B of the encoder need to be necessarily connected to the interrupt pins of the Arduino as the library uses interrupts to track the rotation. An Arduino Uno, Nano, or Pro version has only two hardware interrupt pins, viz, pins 2 and 3. The encoder will not work if connected to any other digital input pin.

This library should be downloaded from *https://github.com/gfvalvo/NewEncoder*. After unzipping the downloaded file, the NewEncoder-master folder should be copied to the ...\Documents\Arduino\Libraries folder.

A 4-digit, 7-segment display is used for displaying the time. A display module based on TM1637 IC has been used as it requires only two pins on an Arduino for displaying the digits. The clock pin of the module is connected to pin 4 while the data pin is connected to pin 5 of the Arduino.

The module uses TM1637 library by Avishay Orpaz. This library can be installed using the 'Manage Libraries' option under the 'Sketch/Include Library' menu of Arduino IDE.

Two momentary press switches (S1 and S2) are used as Start and Stop buttons. One pin of the Start button is connected to pin 6 while the second is grounded. Similarly, the Stop button is connected to pin 7 of the Arduino.

There is no need to use pull-up resistors as the internal pull-ups are activated in software. The button press is handled using OneButton library by Matthias Hertel. It can also be installed using 'Manage Libraries' option under 'Sketch/Include Library' menu of the IDE.

A magnetic door sensor switch is fixed on the upper edge of the door as shown in Fig. 8. The magnet section is connected to the door whereas the reed switch part is connected to the box. One wire of the reed switch is connected to pin D12 of Arduino while the second wire is grounded. The mode of D12 pin is set as INPUT_PULLUP so that the internal pull-up resistor of pin D12 is activated.

A 5V buzzer is connected to pin D11 of Arduino to give warning signals.

A cardboard's lid for the box serves as a cover for the chokes and the front edge serves as the control panel. The circuit is placed inside this lid as shown in Fig. 9.

A 5V relay is connected to pin D10 of Arduino via a 2N2222 NPN transistor whose base is connected to 18k resistor to limit the base current. The

EFY Note

The source code of this project is available for free download at *source.efymag.com*